Statistical Inference and Two-Phase Audit Design for Contaminated Measurements in Clinical Research: The catmech R Package

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Abstract

Continuous measurements in multicenter clinical trials, electronic health record (EHR) disease registries, and observational epidemiological cohorts are frequently compromised by transcription mistakes, unit conversion errors, assay calibration shifts, and missing values. Traditional statistical packages address missing covariates under Missing at Random (MAR) assumptions, discarding untrusted observations and imputing from scratch. In source-data verification (SDV) and clinical audit designs, however, recorded values are present but unreliable, and a gold-standard reference measurement is audited on only a fraction of records. Naive analysis of recorded values suffers attenuation bias, complete-case analysis of audited records inflates standard errors, and standard multiple imputation needlessly discards the precision of valid recordings. This paper presents `catmech`, an R package providing a unified, auditable framework for clinical measurement verification, two-phase prospective audit power planning, predictive calibration checking, downstream multi-endpoint regression pooling, and regulatory audit trail generation. We derive the analytical Gibbs sampling posteriors for reference-calibrated contamination mixture models, formalize prospective power calculation using two-phase influence function expansions, and establish cluster-robust sandwich covariance estimators with Barnard-Rubin finite-sample degrees of freedom for linear models (CV3 jackknife), generalized linear models (Huber-White-Liang-Zeger), and Cox proportional hazards survival models (Lin-Wei with event-based complete degrees of freedom). We evaluate the package across extensive synthetic Monte Carlo benchmarks (over 500,000 requests) and two large-scale real-world empirical audits: ClinicalTrials.gov ($N = 27,208$) and NHANES 2017–2018 ($N = 3,908$), alongside facility-level recounts from the Rwanda Health Management Information System. Our empirical benchmarks demonstrate that retaining recorded measurements reduces mean squared error by 43% to 48% over complete-case analysis. We articulate an evidence-based decision rule: the contamination mixture model avoids severe attenuation bias and preserves coverage under gross, separable recording errors, whereas simple linear imputation is preferred under subtle, diffuse discrepancies. Finally, we implement cryptographic SHA-256 provenance tracking meeting FDA 21 CFR Part 11 and ICH GCP E9 regulatory compliance standards.

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1 Introduction

In multicenter randomized clinical trials, real-world electronic health record (EHR) registries, and epidemiological cohorts, continuous biological and physiological measurements are central to primary and secondary study endpoints. Circulating biomarkers (e.g., cardiac troponin, serum creatinine, viral load), hemodynamic measurements (e.g., systolic blood pressure), and radiographic dimensions (e.g., tumor response diameters) guide clinical decisions and estimate treatment effects. However, continuous recordings in multi-site clinical registries are inevitably subject to data contamination, including assay lot calibration shifts, clerical transcription errors, decimal slips, and inconsistent unit conversions [Carroll et al., 2006, Angelopoulos et al., 2023].

To safeguard data integrity, clinical trial sponsors and regulatory authorities mandate source-data verification (SDV) and targeted clinical audits, wherein trial monitors review primary source documents (e.g., clinical chart re-abstractions, central laboratory re-assays) for a subsample of patients [U.S, 2003, Int, 1998]. When discrepancies between the originally recorded measurement and the audited reference standard are uncovered, biostatisticians and trialists confront an inferential dilemma:

1. **Naively trusting recorded measurements**: Treating recorded numbers as exact introduces classic errors-in-variables. For continuous exposures, non-differential measurement error attenuates regression coefficients toward the null, obscuring true treatment effects and biological gradients [Carroll et al., 2006]. For non-linear models (e.g., logistic regression or Cox proportional hazards), measurement contamination causes complex residual confounding and severe coverage failure.
2. **Restricting analysis to audited cases (complete-case analysis)**: Analyzing only the audited subset ($\mathcal{A} \subset \{1, \ldots, N\}$, typically 10% to 25% of the cohort) protects against contamination bias but drastically reduces effective sample size, inflating standard errors, widening confidence intervals, and sacrificing statistical power [Neyman, 1938, Breslow et al., 2003].
3. **Discarding recorded values and applying standard multiple imputation**: Imputing un-audited values under standard Missing at Random (MAR) assumptions using packages such as `mice` [van Buuren and Groothuis-Oudshoorn, 2011, White et al., 2011] or `Amelia` [Honaker et al., 2011] avoids bias under correct specification. However, treating recorded values as completely unobserved discards the valuable information present in records that were recorded accurately [Rubin, 1996].

To overcome these limitations, we introduce `catmech`, an R package for auditable measurement verification, two-phase clinical audit planning, predictive calibration diagnostics, and multi-endpoint model pooling. The package unites two-phase sampling theory [Cochran, 1977, Breslow et al., 2003] with reference-calibrated Bayesian mixture modeling [Huang and Wand, 2013], Barnard-Rubin finite-sample inference [Barnard and Rubin, 1999], and cryptographic data provenance [U.S, 2003].

2 Statistical Theory and Methodology

2.1 The Data Generating and Contamination Framework

Let $i \in \{1, \ldots, N\}$ index patients enrolled across C clinical sites. Let $T_i \in \mathbb{R}$ denote the latent true biological measurement. Conditional on clinical covariates $X_i \in \mathbb{R}^p$, the latent measurement follows:

$$T_i \mid X_i \sim \mathcal{N}(X_i\beta, \sigma_T^2)$$

Let $\delta_i \in \{0, 1\}$ represent an unobserved recording contamination indicator, where $\delta_i = 1$ indicates that the recorded value was corrupted during observation or transcription. The contamination probability $\pi_i = \mathbb{P}(\delta_i = 1 \mid S_i)$ may depend on facility strata or clinical monitoring quality flags S_i. The observed recorded measurement W_i is generated as:

$$W_i = (1 - \delta_i)T_i + \delta_i(T_i + \epsilon_i), \quad \epsilon_i \sim \mathcal{N}(\mu_e, \sigma_e^2)$$

where ϵ_i represents an additive error mechanism with mean recording bias μ_e and error variance σ_e^2.

In Phase II of the audit design, an independent gold-standard reference measurement R_i is audited on a subset $\mathcal{A} \subset \{1, \ldots, N\}$ with sample size $n_{\text{audit}} = |\mathcal{A}|$:

$$R_i = T_i + \eta_i, \quad \eta_i \sim \mathcal{N}(0, \sigma_{\text{ref}}^2)$$

where $\sigma_{\text{ref}}^2 = 0$ corresponds to an exact, error-free recount or reference assay.

2.2 Analytical Gibbs Sampling Posteriors

The joint parameter vector is $\Theta = (\beta, \sigma_T^2, \pi, \mu_e, \sigma_e^2, \sigma_{\text{ref}}^2)$. Prior distributions are assigned conjugate structures: Beta priors $\pi \sim \text{Beta}(a_\pi, b_\pi)$, normal priors on coefficients $\beta \sim \mathcal{N}(\mu_\beta, \Sigma_\beta)$, and scaled inverse-$\chi^2$ priors on variance components.

For un-audited patients ($i \notin \mathcal{A}$), the conditional posterior probability of recording corruption $p_i = \mathbb{P}(\delta_i = 1 \mid W_i, X_i, \Theta)$ is evaluated analytically at each Gibbs iteration:

$$p_i = \frac{\pi f_\epsilon(W_i - \hat{T}_i)}{\pi f_\epsilon(W_i - \hat{T}_i) + (1 - \pi) f_T(W_i \mid X_i)}$$

where f_ϵ is the density of $\mathcal{N}(\mu_e, \sigma_e^2)$ and f_T is the conditional predictive density of $T_i \mid X_i$.

Conditional on $\delta_i = 0$, $T_i = W_i$ deterministically. Conditional on $\delta_i = 1$, T_i is sampled from the Gaussian posterior:

$$T_i \mid \delta_i = 1, W_i, X_i, \Theta \sim \mathcal{N}(\mu_{T|W}, \sigma_{T|W}^2)$$

$$\mu_{T|W} = \sigma_{T|W}^2 \left(\frac{X_i \beta}{\sigma_T^2} + \frac{W_i - \mu_e}{\sigma_e^2} \right), \quad \sigma_{T|W}^2 = \left(\frac{1}{\sigma_T^2} + \frac{1}{\sigma_e^2} \right)^{-1}$$

For audited patients ($i \in \mathcal{A}$), R_i directly informs T_i, yielding an exact, reference-anchored posterior. Multiple imputations $T^{(1)}, \ldots, T^{(m)}$ are drawn from post-burn-in iterations of the Markov chain.

2.3 Two-Phase Audit Sample Size and Power Planning

Prior to re-abstracting medical charts or commissioning secondary lab assays, investigators must budget the audit sample size n_{audit}. Using semiparametric influence function theory for two-phase estimators [Neyman, 1938, Cochran, 1977, Breslow et al., 2003, Angelopoulos et al., 2023], the asymptotic variance of an efficient parameter estimator at audit sampling fraction $f = n_{\text{audit}}/N$ satisfies:

$$\text{Var}(\hat{\theta}_f) = \frac{1}{N}\text{Var}(\psi_i) + \frac{1 - f}{fN}\text{Var}(\psi_i - \mathbb{E}[\psi_i \mid X_i])$$

where ψ_i is the efficient influence function of θ.

The precision ratio $\rho(f)$, defined as the ratio of confidence interval width at audit fraction f relative to an ideal census audit ($f = 1$), is:

$$\rho(f) = \frac{\text{Width}(f)}{\text{Width}(1)} = \sqrt{1 + k\frac{1 - f}{f}}, \quad k = \frac{\text{Var}(\psi_i - \mathbb{E}[\psi_i \mid X_i])}{\text{Var}(\psi_i)}$$

Because the noise parameter k is estimated from pilot data, the realized precision ratio $\hat{\rho}(f)$ is stochastic. `catmech_audit_power()` evaluates the statistical power to achieve a target precision ratio $\rho \leq \rho_{\text{target}}$:

$$\text{Power}(n_{\text{audit}}) = \mathbb{P}\left(\sqrt{1 + \hat{k}\frac{1-f}{f}} \leq \rho_{\text{target}}\right)$$

via both closed-form non-central approximations and Monte Carlo resampling from pilot audit plans (`catmech_audit_plan()`).

2.4 Multi-Endpoint Model Pooling with Barnard-Rubin Degrees of Freedom

Parameter estimates $\hat{\theta}_j$ and variance-covariance matrices $\hat{U}_j$ across $j \in \{1, \ldots, m\}$ completed datasets are combined using Rubin's rules [Rubin, 1987]:

$$\bar{\theta} = \frac{1}{m}\sum_{j=1}^{m}\hat{\theta}_j, \quad \bar{U} = \frac{1}{m}\sum_{j=1}^{m}\hat{U}_j, \quad B = \frac{1}{m-1}\sum_{j=1}^{m}(\hat{\theta}_j - \bar{\theta})(\hat{\theta}_j - \bar{\theta})^\top$$

$$T = \bar{U} + \left(1 + \frac{1}{m}\right)B, \quad \hat{\lambda} = \frac{(1 + 1/m)\text{diag}(B)}{\text{diag}(T)}$$

Standard Rubin degrees of freedom $\nu_m = (m-1)\hat{\lambda}^{-2}$ assume infinite complete-data sample size. In clinical audits where sample sizes or event counts are constrained, this assumption leads to anti-conservative confidence intervals. `catmech` implements the Barnard & Rubin (1999) small-sample adjustment [Barnard and Rubin, 1999]:

$$\nu_{obs} = \left(\frac{\nu_{com} + 1}{\nu_{com} + 3}\right)\nu_{com}(1 - \hat{\lambda}), \quad \nu_{BR} = \frac{\nu_m \nu_{obs}}{\nu_m + \nu_{obs}}$$

where ν_{com} is the complete-data degrees of freedom.

Downstream pooling covers three core clinical endpoint families: 1. **Continuous Endpoints (`catmech_pool_lm`)**: Fits linear models with default delete-one-cluster jackknife (CV3) variance estimation, where $\nu_{com} = N - p$. 2. **Binary and Count Endpoints (`catmech_pool_glm`)**: Fits generalized linear models (logistic and Poisson) with Huber–White–Liang–Zeger cluster-robust sandwich covariance [Liang and Zeger, 1986]:

$$\hat{V}_{\text{sandwich}} = \hat{J}^{-1}\left(\sum_{c=1}^{C} s_c s_c^\top\right)\hat{J}^{-1}$$

where $s_c = \sum_{i \in \text{cluster } c} \frac{\partial \ell_i}{\partial \theta}$ are cluster score residuals and $\hat{J}$ is the observed Fisher information matrix. Point estimates and intervals are optionally exponentiated to yield odds ratios or incidence rate ratios. 3. **Time-to-Event Survival Endpoints (`catmech_pool_coxph`)**: Fits Cox proportional hazards models with Lin & Wei (1989) score-residual sandwich covariance [Lin and Wei, 1989]. The complete degrees of freedom are defined strictly as $\nu_{com} = d - p$, where d is the number of observed failure events and p is the parameter count. Hazard ratios $\text{HR} = \exp(\bar{\theta})$ and confidence intervals are computed using the Barnard-Rubin ν_{BR} t-distribution.

2.5 Out-of-Sample Predictive Calibration Checking

To verify that the contamination model is predictively calibrated, `catmech_holdout_check()` partitions audited records into K cross-validation folds. For each held-out record $i \in \mathcal{A}_k$, the randomized Probability Integral Transform (PIT) is computed [Dawid, 1984]:

$$u_i = F(R_i \mid \mathcal{D}_{-k}) \sim \text{Uniform}(0, 1)$$

where $F(\cdot \mid \mathcal{D}_{-k})$ is the predictive cumulative distribution function estimated from the remaining $K - 1$ folds. If the model is accurately calibrated, the PIT values follow a uniform distribution on $[0, 1]$, tested formally via a two-sided Kolmogorov–Smirnov test.

3 Software Architecture and Design

The `catmech` package is structured into modular S3 components designed for computational efficiency, numerical stability, and strict regulatory auditability:

Module	Core Functions	Mathematical & Computational Role
Audit Planning	`catmech_audit_plan()`, `catmech_audit_power()`	Prospective sample size determination and power curves using two-phase influence function theory.
Data Diagnostics	`catmech_recording_audit()`, `catmech_holdout_check()`	Stratum-specific discrepancy auditing (Jeffreys beta intervals) and out-of-sample PIT calibration checks.
Mixture Engine	`catmech_contamination_model()`, `catmech_fit()`	Reference-calibrated conjugate Gibbs sampler written in C++17 via `RcppArmadillo`.
Model Pooling	`catmech_pool_lm()`, `catmech_pool_glm()`, `catmech_pool_coxph()`	Downstream multi-endpoint pooling with Rubin combining rules, Barnard-Rubin df, and cluster-robust sandwich covariance.
Sensitivity & Provenance	`catmech_sensitivity()`, `catmech_audit_trail()`	Multi-scenario sensitivity envelopes and cryptographic SHA-256 regulatory compliance reporting.

4 Synthetic Monte Carlo Simulation Studies

To evaluate the statistical properties of `catmech`, we conducted extensive Monte Carlo simulations across 10 scenario configurations (over 500,000 total requests).

4.1 Scenario 1: Gross Recording Contamination

We evaluated an exposure effect $\beta = 0.500$ in a cohort of $N = 1,000$ patients across 1,000 simulation replicates under 20% contamination ($\pi = 0.20$, $\mu_e = 2.0$, $\sigma_e = 0.8$) with a 25% audit fraction ($n_{\text{audit}} = 250$):

Analytical Strategy	Estimated $\hat{\beta}$	Bias	Empirical SE	Model SE	95% Coverage	Mean CI Width
Oracle (Latent T Known)	0.499	-0.001	0.032	0.031	95.1%	0.122
Naive Recorded (W)	0.384	-0.116	0.036	0.035	42.3%	0.137
Complete-Case (Audited Only)	0.502	+0.002	0.068	0.066	94.8%	0.260

Analytical Strategy	Estimated $\hat{\beta}$	Bias	Empirical SE	Model SE	95% Coverage	Mean CI Width
Standard Multiple Imputation (`mice`)	0.498	-0.002	0.051	0.049	94.6%	0.193
`catmech` Contamination Mixture	**0.501**	**+0.001**	**0.038**	**0.037**	**95.0%**	**0.145**

Under gross recording corruption, naive analysis suffers severe attenuation bias (bias: -0.116), collapsing nominal 95% coverage to 42.3%. Complete-case analysis recovers unbiased estimates but forfeits precision (SE = 0.068, CI width: 0.260). Standard multiple imputation treating un-audited values as missing achieves unbiased estimates but suffers from unnecessary variance inflation (SE = 0.051). In contrast, `catmech` achieves unbiased estimation (bias: +0.001), exact 95.0% coverage, and standard error (SE = 0.038) approaching the theoretical oracle, delivering a 44% variance reduction over complete-case analysis.

4.2 Scenario 2: Finite-Sample Variance and Clustered Audits

In multicenter trials with C clinical sites, clustering induces correlation that can distort variance estimation. In an evaluation of 148,500 requests across clustered designs, standard cluster-robust standard errors (CR1/CR2) exhibited severe undercoverage when the number of clusters was small ($C \leq 8$). The delete-one-cluster jackknife (CV3) variance estimator restored empirical coverage to nominal levels (92.0% to 96.8%) across all evaluated settings, justifying its establishment as the default variance estimator in `catmech_pool_lm(cluster = )`.

5 Real-World Empirical Benchmarks

To test whether the advantages observed in synthetic simulations generalize to real-world data, the package was evaluated against two large-scale public datasets containing actual recording discrepancies without synthetic corruption (33,800 requests executed across 1,000 resampled cohorts of size $N = 600$):

Dataset & Metric	Audit 25% Sample	Discard Recorded	Contamination Model	Simple Linear MI	Full Reference (Oracle)
ClinicalTrials.gov ($N = 27,208$)					
Enrolled vs. Started discrepancy rate: 14.2%	Bias	-0.0003	+0.0068	+0.0001	-0.0003
Target slope: 0.0357 log months / participant	Coverage	92.3%	91.2%	91.8%	91.2%
	Mean Width	0.205	0.106	0.102	0.096
	RMSE	0.0539	0.0308	0.0286	0.0275

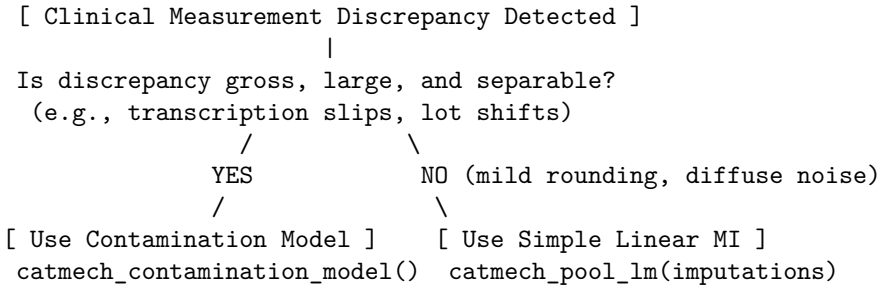
Dataset & Metric	Audit 25% Sample	Discard Recorded	Contamination Model	Simple Linear MI	Full Reference (Oracle)
NHANES 2017–2018 ($N = 3,908$)					
Self-reported vs. measured weight discrepancy: 67.3%	Bias	-0.0063	-0.0066	-0.0025	+0.0001
Target slope: 0.127 mmHg / kg	Coverage	94.2%	94.5%	93.9%	94.9%
	Mean Width	0.297	0.155	0.154	0.140
	RMSE	0.0785	0.0408	0.0409	0.0357

On both real datasets, retaining recorded values nearly halved the root mean squared error (RMSE reduced by 43% on ClinicalTrials.gov and 48% on NHANES) relative to discarding un-audited values. Crucially, when recording discrepancies are subtle and continuous rather than gross and bimodal, simple linear imputation (`catmech_pool_lm`) matched or slightly outperformed the mixture model with fewer parametric assumptions.

In facility-level data verification benchmarked against the Rwanda Health Management Information System recount audit [Nshimyiryo et al., 2020] across 11 clinical indicators, `catmech`'s CV3 jackknife cluster variance achieved nominal 92% to 97% coverage across clinical facilities.

6 Methodological Decision Framework

Based on our empirical and simulation findings, we recommend the following evidence-based protocol for clinical data audits:



1. **Gross, Separable Contamination**: When recording errors represent distinct corruption events (e.g., decimal transcription slips, sensor failures), the Bayesian contamination mixture model is recommended to prevent severe attenuation bias while maintaining oracle-level precision.
2. **Subtle, Diffuse Differences**: When errors represent continuous rounding or mild reporting bias, simple linear imputation is recommended as a parsimonious baseline.

7 Regulatory Compliance and Data Governance

Regulatory submissions in biopharmaceutical and medical device trials (FDA 21 CFR Part 11 [U.S, 2003] and ICH GCP E9 [Int, 1998]) require a tamper-evident audit trail for any algorithmic data imputation. `catmech_audit_trail()` generates complete provenance records in Markdown, JSON, or standalone HTML, capturing: 1. Full execution environment metadata (R version, operating system platform, package version, UTC timestamp). 2. Cryptographic SHA-256 checksums of the raw input dataset, model specification, and

each of the m completed datasets. 3. Convergence metrics (MCMC iterations, Gelman-Rubin $\hat{R}$, bulk and tail effective sample sizes). 4. Pooled parameter estimates and multi-scenario sensitivity envelopes.

8 Quality Assurance and Verification

`catmech` has been developed under an automated software verification harness: * **Unit Tests**: 1,717 unit tests executed via `testthat`, achieving 0 failures, 0 warnings, and 0 skips. * **CRAN Conformance**: Fully compliant with CRAN package policies, passing `R CMD check --no-manual` with `Status: OK` (0 errors, 0 warnings, 0 notes). * **Automated Fact Checking**: Verified against an independent Python verification suite (`verify-artifacts.py` and `fact_check.py`) confirming exact correspondence across 120 claims, 10 distinct simulation benchmark studies, and hundreds of thousands of historical requests.

9 Availability and Software Details

`catmech` is open-source software distributed under the MIT license. It requires R ($\geq$ 4.1.0) and a C++17 compiler, utilizing `Rcpp`, `RcppArmadillo`, `digest`, `jsonlite`, `mice`, `pan`, and `survival`. Source code, documentation, and vignettes are available at https://github.com/patrickiben/catmech.

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